Observe Notifications as CoAP Multicast Responses

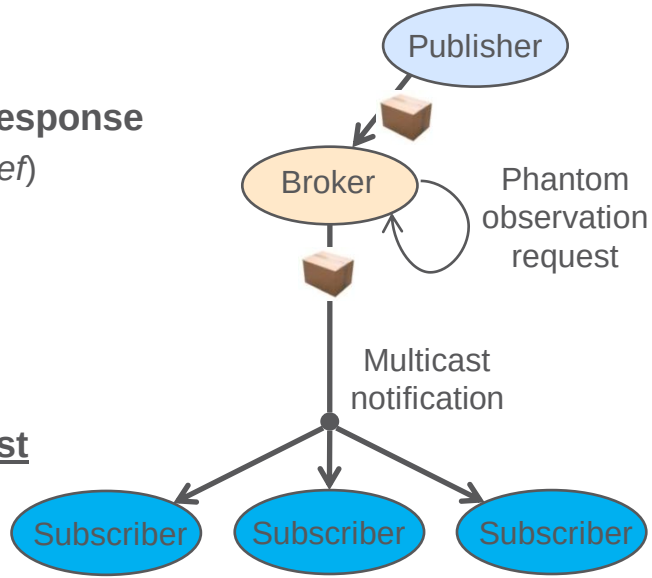
draft-ietf-core-observe-multicast-notifications-15

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Recap

- **Observe notifications (RFC 7641) as multicast responses**
  - Many clients observe the same resource on a server (e.g., pub-sub)
  - Improved performance due to multicast delivery
- **Clients configured by the server, with a 5.03 error informative response**
  - Transport-specific information is provided as CRIs (*draft-ietf-core-href*)
- **All clients in a group observation use the same Token value**
  - The Token space belongs to the group (clients)
  - The group entrusts the server to manage the token space
- **Multicast notifications bound to a Phantom Observation Request**
  - By means of the same Token value for that observation
- **Group OSCORE to protect multicast notifications**
  - The server aligns all clients of an observation on a same *external_aad*
  - All notifications for a resource are protected using that *external_aad*



Example in a pub-sub scenario

Since IETF 124 (November 2025)

- **Two updates to this document**
  - Version -14 in April 2026
  - Version -15 in July 2026

- **Updates to the corresponding *draft-ietf-core-multicast-notifications-proxy***
  - Now in version -02; not covered today

Updates from -13 to -14

- **Editorial**
  - Updates references; minor fixes in examples
  - Removed unnecessary normative language (behavior inherited from other documents)

- **Note on the HKDF Algorithm used in Group OSCORE**
  - Indicated by the corresponding HMAC Algorithm, like in other OSCORE-related specs

- **In Appendix D on using Deterministic Requests [1] as phantom requests**
  - Clarified pre-conditions: support for Deterministic Requests as in [1]
  - Discussed advantages:
    - Clients can start from a smaller, unprotected version of the phantom observation request
    - No need to verify that the phantom observation request is generated by the server
  - If the OSCORE group uses Deterministic Requests and clients are known to support those, then servers are encouraged to use Deterministic Requests as phantom requests

[1] https://datatracker.ietf.org/doc/draft-ietf-core-cacheable-oscore/

Updates from -13 to -14

- **Section 9 – When using Group OSCORE, expected roles in the group:**
  - Server: join the group (also) with the roles of Requester and Responder
  - Client: join the group with any permitted role or combination of roles
    - If Group OSCORE also protects the original observation request, then also as a Requester

- **Section 8.1 – Generalized semantics of a new CoAP option**
  - Old name: “Multicast-Response-Feedback-Divider”
  - New semantics: *a sender endpoint elicits a stochastic reaction and thereby a feedback from the message recipient(s) that would otherwise not react.*
  - Unchanged use in this document: perform rough counting of active observer clients
  - New name: “Feedback-Divider”

Updates from -14 to -15

- **Editorial fixes, updated references, etc.**

- **Fixes in CDDL notations**
  - Section 4.2 – Payload of the informative response: `*0 => array*` → `*0 => [1* tp_info_element]*`
  - Section 4.2.1.1 – 'tp_info' array for CoAP over UDP: added definition of `*CRI-no-local*`

- **Section 9.2.1 on using Group OSCORE to protect the phantom request**
  - Clarification: the group mode of Group OSCORE is used, with the exception discussed in Appendix D for the Deterministic Requests [1]

[1] https://datatracker.ietf.org/doc/draft-ietf-core-cacheable-oscore/

Updates from -14 to -15

- **Section 4.2 – 5.03 informative response**
  - ‘last_notif’ – Latest multicast notification sent by the server
  - Update: ‘last_notif’ SHOULD be included
  - Avoid unnecessary delays for the client in obtaining a first multicast notification and possibly populating its cache

```
informative_response_payload = {
 ? 0 => [1* tp_info_element],  ; 'tp_info'     (transport-specific
                               ;                information)
 ? 1 => bstr,                  ; 'ph_req'      (transport-independent
                               ;                information)
 ? 2 => bstr,                  ; 'last_notif'  (transport-independent
                               ;                information)
 ? 3 => uint,                  ; 'next_not_before'
 ? 4 => ~time                  ; 'ending'
 * (int / tstr) => any
}

tp_info_element = any
```

- **Age of multicast notifications**
  - If sent on the wire as stand-alone messages → Nothing special, look at the Max-Age Option
  - If included as ‘last_notif’ in an informative response → Their indicated age is misleading

- **When the latest multicast notification reaches its maximum age at the server …**
  - Section 4.3 – The server MUST compose and send a new multicast notification
  - Section 5.2 – Consideration for clients about age of notifications taken from ‘last_notif’

Updates from -14 to -15

- **Section 4.2 – 5.03 informative response**
  - ‘tp_info’ – transport-specific information about the multicast notifications
- **‘tp_info’ MAY be omitted in some setups**
  - The server could know that transport-specific information is available to clients in other ways
  - E.g., Group OSCORE is used, but notifications are ultimately delivered to observer clients over a unicast transport, e.g., through a proxy [2]
- **Similarly, when using CoAP over UDP …**
  - ‘tpi_details’ MAY be omitted from ‘tp_info’
- **Otherwise they MUST be included**

```
informative_response_payload = {
 ? 0 => [1* tp_info_element],  ; 'tp_info'     (transport-specific
                               ;                information)
 ? 1 => bstr,                  ; 'ph_req'      (transport-independent
                               ;                information)
 ? 2 => bstr,                  ; 'last_notif'  (transport-independent
                               ;                information)
 ? 3 => uint,                  ; 'next_not_before'
 ? 4 => ~time                  ; 'ending'
 * (int / tstr) => any
}

tp_info_element = any


 tp_info = [
    tpi_server: CRI-no-local,  ; Addressing information of the server
   ? tpi_details               ; Further information about the request
 ]

 tpi_details = (
   + elements  ; The number, format, and encoding of the elements
               ; depend on the scheme-id and authority of the CRI
               ; specified by tpi_server
 )

 CRI-no-local = [
   scheme-id,
   authority
 ]
```

[2] https://datatracker.ietf.org/doc/draft-ietf-core-multicast-notifications-proxy/

Updates from -14 to -15

- **New Section 13 “Operational Considerations”**
  - Section 13.1 – Logging of events at the server
  - Section 13.2 – Management and distribution of keying material
  - Section 13.3 – Access control

Implementation

- **Eclipse Californium (Java) – Adrian Kastrati**
  - https://github.com/adriankastrati/californium-thesis-work/tree/multicast-notifications
- **Tested with 3 observer clients and one server**
- **Tested without security**
- **Tested with security**
  - OSCORE to protect regular Observe requests and 5.03 informative responses
  - Group OSCORE to protect the phantom request and the multicast notifications

Next steps

- **This version should be ready for WG Last Call**

- **Normative references**
  - 4 Internet Drafts are approved for publication and with the RFC Editor
  - *draft-ietf-cbor-edn-literals* completed the second WG Last Call
    - Informational in the Datatracker although intended as Standards Track

Thank you!

Comments/questions?

https://github.com/core-wg/observe-multicast-notifications